

Reinforcement Learning

Introduction

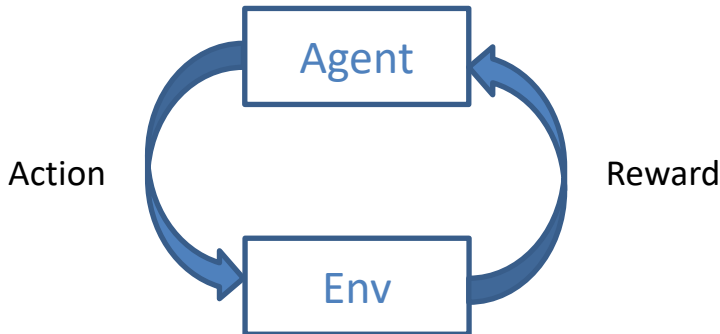
AI plays Flappy Bird

AI plays DOTA 2

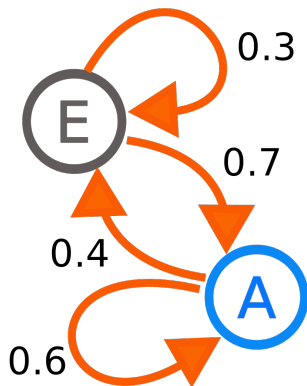


What is Reinforcement Learning

Reinforcement learning (RL) is an area of machine learning that focuses on how **agent** might act in an **environment** in order to maximize some given reward.



Markov Decision Process



Markov chain

Markov Decision Process

Markov Decision Process (MDP) is a process of the agent observing the environment output consisting of a reward and the next state, and then acting upon that.

Components of an MDP:

- Agent
- Environment
- State
- Action
- Reward

Markov Decision Process

In an MDP, we have a set of states $\mathcal{S}$, a set of actions $\mathcal{A}$, and a set of rewards $\mathcal{R}$. We'll assume that each of these sets has a finite number of elements.

At each time step $t = 0, 1, 2, \dots$, the agent receives some representation of the environment's state $S_t \in \mathcal{S}$. Based on this state, the agent selects an action $A_t \in \mathcal{A}$. This gives us the state-action pair (S_t, A_t) .

Time is then incremented to the next time step $t + 1$, and the environment is transitioned to a new state $S_{t+1} \in \mathcal{S}$. At this time, the agent receives a numerical reward $R_{t+1} \in \mathcal{R}$ for the action A_t taken from state S_t .

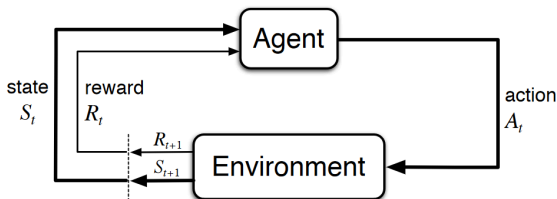
We can think of the process of receiving a reward as an arbitrary function f that maps state-action pairs to rewards. At each time t , we have

$$f(S_t, A_t) = R_{t+1}.$$

The trajectory representing the sequential process of selecting an action from a state, transitioning to a new state, and receiving a reward can be represented as

$$S_0, A_0, R_1, S_1, A_1, R_2, S_2, A_2, R_3, \dots$$

Markov Decision Process



1. At time t , the environment is in state S_t .
2. The agent observes the current state and selects action A_t .
3. The environment transitions to state S_{t+1} and grants the agent reward R_{t+1} .
4. This process then starts over for the next time step, $t + 1$.
 - Note, $t + 1$ is no longer in the future, but is now the present. When we cross the dotted line on the bottom left, the diagram shows $t + 1$ transforming into the current time step t so that S_{t+1} and R_{t+1} are now S_t and R_t .